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JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 108, NO. B4, 2181,
doi:10.1029/2002JB001909, 2003

Paleomagnetism and magnetostratigraphy of the lower Glen Canyon and upper Chinle Groups, Jurassic-Triassic of northern Arizona and northeast Utah

Roberto S. Molina-Garza
Centro de Geociencias, UNAM Campus Juriquilla,
Querétaro, México

John W. Geissman
Department of Earth and Planetary Sciences, University of New Mexico,
Albuquerque, New Mexico, USA

Spencer G. Lucas
New Mexico Museum of Natural History and Science,
Albuquerque, New Mexico, USA

Abstract

[1] Twenty-eight selected sites (individual beds) in the Moenave Formation at the Echo Cliffs, northern Arizona, strata give a Hettangian paleomagnetic pole at 63.7° N, 59.7°E (*dp* = 2.6°, *dm* = 5.1°). The Wingate Sandstone and Rock Point Formation at Comb Ridge, southeast Utah, provide a Rhaetian paleopole at 57.4°N, 56.6°E (*N* = 16 sites; *dp* = 3.4, *dm* = 6.5). High unblocking temperatures (>600°C), high coercivity, and data analyses indicate that the characteristic magnetization is primarily a chemical remanence residing in hematite. The Hettangian and Rhaetian poles are statistically indistinguishable (at 95% confidence), they resemble existing data for the Glen Canyon Group, and they provide further validation to the J1 cusp of the North American apparent pole wander path (APWP). The red siltstone and upper members of the Chinle Group, on the south flank of the Uinta Mountains, northern Utah, define a Rhaetian pole at 51.6°N, 70.9°E (*N* = 20 sites; *dp* = 3.5°, *dm* = 6.9°). The Gartra and upper members of the Chinle Group in the north flank of the Uinta Mountains, give paleopoles at 52.0°N, 100.3°E (*N* = 6 sites; *dp* = 5.4°, *dm* = 10.5°) and 50.9°N, 50.1°E (*N* = 5 sites; *dp* = 8.8°, *dm* = 17.5°), respectively. These data

indicate no significant rotation of the Uinta Mountains with respect to the craton. In total, data for the plateau and its bordering region of Cenozoic uplifts support estimates of small rotation of the plateau and provide evidence against the hypothesis of a Late Triassic standstill of the North American APWP. Our magnetostratigraphic results are consistent with lithographic and biostratigraphic data that place the Triassic-Jurassic boundary within the Dinosaur Canyon Member of the Moenave Formation, not at a regional hiatus.

Received 1 April 2002; revised 7 October 2002; accepted 12 December 2002; published 2 April 2003.

Keywords: paleomagnetism, Triassic, Jurassic, magnetostratigraphy, Colorado Plateau.

Index Terms: 1520 Geomagnetism and Paleomagnetism: Magnetostratigraphy; 1525 Geomagnetism and Paleomagnetism: Paleomagnetism applied to tectonics (regional, global); 1527 Geomagnetism and Paleomagnetism: Paleomagnetism applied to geologic processes.

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Citation: Molina-Garza, R. S., J. W. Geissman, and S. G. Lucas, Paleomagnetism and magnetostratigraphy of the lower Glen Canyon and upper Chinle Groups, Jurassic-Triassic of northern Arizona and northeast Utah, *J. Geophys. Res.*, *108*(B4), 2181, doi:10.1029/2002JB001909, 2003.

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